DISCRETE STRUCTURES

CMSC250

Chapter 19: Advanced Set Operations

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Power sets, products, partitions, and identities



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1 Why advanced set constructions matter

A set can contain numbers, other sets, ordered pairs, or whole collections of objects. **Power sets** collect subsets, **Cartesian products** collect ordered tuples, and **partitions** divide a set into nonoverlapping pieces. Each construction has its own membership rule.

2 Power sets and the subset test

By definition, $\mathcal{P}(A) = \{X : X \subseteq A\}$. Thus $X \in \mathcal{P}(A)$ is equivalent to $X \subseteq A$. If $A = \{a, b, c\}$, then $\mathcal{P}(A)$ contains eight sets: the empty set, three singletons, three two-element subsets, and A itself.

Key idea. When proving $\mathcal{P}(A) \subseteq \mathcal{P}(B)$, begin with an arbitrary *set* $X \in \mathcal{P}(A)$, not an arbitrary number $x \in A$.

Worked example: power-set inclusion

Question. Prove $A \subseteq B$ implies $\mathcal{P}(A) \subseteq \mathcal{P}(B)$.

Proof. Assume $A \subseteq B$. Let $X \in \mathcal{P}(A)$ be arbitrary. By the definition of power set, $X \subseteq A$. Since $A \subseteq B$, transitivity of subset containment gives $X \subseteq B$. Therefore $X \in \mathcal{P}(B)$. Thus $\mathcal{P}(A) \subseteq \mathcal{P}(B)$. ■

Worked example: the converse of power-set inclusion

Question. Prove $\mathcal{P}(A) \subseteq \mathcal{P}(B)$ implies $A \subseteq B$.

Proof. Assume $\mathcal{P}(A) \subseteq \mathcal{P}(B)$. Since $A \subseteq A$, we have $A \in \mathcal{P}(A)$. By the assumption, $A \in \mathcal{P}(B)$. Hence $A \subseteq B$, as required. ■

3 Power sets and intersections

Worked example: an identity involving power sets

Question. Prove $\mathcal{P}(A \cap B) = \mathcal{P}(A) \cap \mathcal{P}(B)$.

Proof. Let X be an arbitrary set. Then

$$\begin{aligned} X \in \mathcal{P}(A \cap B) &\iff X \subseteq A \cap B \\ &\iff (X \subseteq A) \wedge (X \subseteq B) \\ &\iff X \in \mathcal{P}(A) \cap \mathcal{P}(B). \end{aligned}$$

The middle equivalence holds because every member of X lies in $A \cap B$ exactly when every member lies in both A and B. Thus the two power sets are equal. ■

4 Why power sets do not distribute over unions

Worked example: a tempting but false identity

Question. Is $\mathcal{P}(A \cup B) = \mathcal{P}(A) \cup \mathcal{P}(B)$ always true?

Solution. No. Let $A = \{1\}$ and $B = \{2\}$. Then $\{1, 2\} \in \mathcal{P}(A \cup B)$, but $\{1, 2\}$ is not a subset of A and not a subset of B. Hence it is absent from $\mathcal{P}(A) \cup \mathcal{P}(B)$. The proposed equality fails.

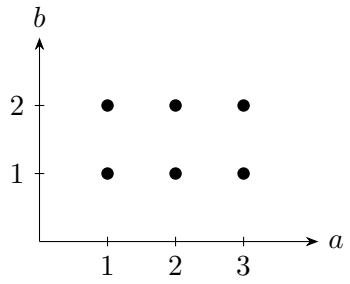
5 Cartesian products and ordered pairs

$A \times B = \{(a, b) : a \in A,\ b \in B\}$. Ordered pairs satisfy $(a, b) = (c, d)$ iff $a = c$ and $b = d$. The first coordinate comes from A, the second from B.

Worked example: draw a finite Cartesian product

Question. Let $A = \{1, 2, 3\}$ and $B = \{1, 2\}$. List and graph $A \times B$.

Solution. The six points are $(1, 1), (1, 2), (2, 1), (2, 2), (3, 1), (3, 2)$.



The graph makes $|A \times B| = |A||B| = 3 \cdot 2 = 6$ visible.

6 Product identities

Worked example: product distributes over union

Question. Prove $A \times (B \cup C) = (A \times B) \cup (A \times C)$.

Proof. Let (x, y) be an arbitrary ordered pair. Then

$$\begin{aligned}(x, y) \in A \times (B \cup C) &\iff x \in A \wedge (y \in B \vee y \in C) \\ &\iff (x \in A \wedge y \in B) \vee (x \in A \wedge y \in C) \\ &\iff (x, y) \in (A \times B) \cup (A \times C).\end{aligned}$$

Therefore the sets are equal. ■

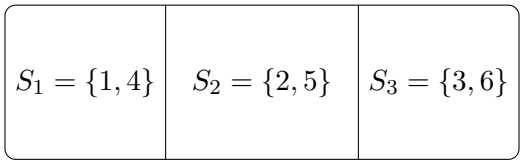
Worked example: product distributes over intersection

Question. Prove $A \times (B \cap C) = (A \times B) \cap (A \times C)$.

Proof. For arbitrary (x, y), membership in the left side means $x \in A$ and $y \in B \cap C$, or equivalently $x \in A$, $y \in B$, and $y \in C$. This holds exactly when (x, y) belongs to both $A \times B$ and $A \times C$. Thus the sets are equal. ■

7 Partitions: breaking a set into pieces

A **partition** of a set S is a collection $\{S_i : i \in I\}$ of subsets such that (1) each S_i is nonempty, (2) distinct blocks are disjoint, and (3) the union of all blocks is S. Each element belongs to exactly one block.



Disjoint blocks covering $S = \{1, 2, 3, 4, 5, 6\}$

Worked example: checking a proposed partition

Question. Does $\{\{1, 2\}, \{3\}, \{4, 5\}\}$ partition $S = \{1, 2, 3, 4, 5\}$?

Solution. Yes. Each block is nonempty, the blocks have no shared elements, and their union is exactly S. By contrast, $\{\{1, 2\}, \{2, 3\}, \{4, 5\}\}$ is not a partition because 2 appears in two blocks.

8 Set identities with three or more sets

For a finite family $A_1, \ldots, A_n$, an element lies in $\bigcup_{i=1}^{n} A_i$ when it belongs to *at least one* A_i; it lies in $\bigcap_{i=1}^{n} A_i$ when it belongs to *every* A_i. For example $\bigcup_{i=1}^{3} \{i, i+1\} = \{1, 2, 3, 4\}$ and $\bigcap_{i=1}^{3} \{i, i+1\} = \varnothing$.

9 Counting elements of finite unions

For finite sets, $|A \cup B| = |A| + |B| - |A \cap B|$. The subtraction prevents counting elements of the overlap twice.

Worked example: students in two clubs

Question. In a class, 18 students join robotics, 15 join mathematics, and 7 join both. How many join at least one?

Solution. Let R and M be the two membership sets. Then $|R \cup M| = 18 + 15 - 7 = 26$. The overlap of seven is counted twice in the initial sum, so it must be subtracted once.

10 Practice questions with complete solutions

Question. 1. Prove $\mathcal{P}(A) = \mathcal{P}(B)$ implies $A = B$.

Solution. Equality gives both $\mathcal{P}(A) \subseteq \mathcal{P}(B)$ and $\mathcal{P}(B) \subseteq \mathcal{P}(A)$. By the converse of power-set inclusion proved earlier, $A \subseteq B$ and $B \subseteq A$. Therefore $A = B$. ■**Question.** 2. Let $A = \{a, b\}$ and $B = \{1, 2, 3\}$. Find $|\mathcal{P}(A \times B)|$.

Solution. The Cartesian product has $2 \cdot 3 = 6$ ordered pairs. Its power set has $2^6 = 64$ elements. **Question.** 3. Does $\{\{1\}, \{2, 3\}, \varnothing\}$ partition $\{1, 2, 3\}$?

Solution. No. Although the nonempty blocks cover the set and do not overlap, a partition cannot contain an empty block.

11 Common mistakes

Do not confuse $X \in \mathcal{P}(A)$ with $X \in A$. Do not assume $\mathcal{P}(A \cup B)$ is the union of the two power sets. Check the order of coordinates in products. A partition requires all three conditions: nonempty blocks, pairwise disjointness, and complete coverage.